Campus Placement & Internship Pipeline – Layered Component Architecture

Presentation Layer

«component»
Student Portal
Resume upload, apply for drives, status dashboard
FR-001, FR-005

«component»
Officer Console
Configure drives and filters, run interview rounds, mark students selected
FR-002, FR-003, FR-004

IApplication
upload resume, apply, read own status

IDriveConfig
set CGPA / backlog filters → saved drive rules

IRoundControl
create queue, mark round cleared

IOfferRequest
mark student selected → send offer letter

Business Layer (stateless services)

«component»
Application & Status Service
Resume parsing, skill tags, application records
FR-001, FR-005

IEligibilityCheck
skills, CGPA, backlogs → eligible / reasons

«component»
Drive Rules & Eligibility Service
Per-drive CGPA / backlog tiers, eligibility decisions
FR-002 ★

IEligibleList
eligible applicants for a drive

«component»
Interview Pipeline Service
Round queues: Aptitude → Technical → HR
FR-003

IRoundOutcome
selected student IDs after final round

«component»
Offer Letter Service
Offer letter PDF generation and email
FR-004

★ NFR-002: services keep no session state, so copies can run side by side for 500 users in < 3 s; active drive rules are held in memory

IDataAccess
read / write records

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Data Layer

«component»
Data Access & Encryption Component
Only gateway to storage: encrypts and decrypts every record (AES-256) | NFR-001

IStorage
encrypted SQL (JDBC)

«component»
Relational Database
Students, drives, rules, rounds, status, offers (stored encrypted)

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Provided interface (ball)
Required interface (socket) – arc faces the consumer
★ Author's focus: FR-002, NFR-002
Ball + socket = assembly connector